Latent Class Analysis of Social Media Influencer Credibility and Purchase Behavior

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Overview

This script was developed as part of a collaborative graduate project (SURV 617). I was responsible for all quantitative analysis and programming. The research question examines which consumer profiles are more likely to make purchasing decisions based on social media influencer recommendations in Bulgaria (n = 376).

The analysis applies: 1. Latent Class Analysis (LCA) to identify unobserved consumer profiles based on perceived influencer credibility (attractiveness, expertise, trustworthiness) 2. Logistic regression to assess how class membership predicts purchase behavior

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Section 1: Load Required Libraries

```
# Core data manipulation and visualization
library(readxl)     # Reading Excel data files
library(dplyr)      # Data wrangling
library(tidyr)      # Reshaping data between wide and long formats
library(ggplot2)    # Data visualization
library(forcats)    # Factor handling
library(stringr)    # String manipulation

# Statistical modeling
library(poLCA)      # Polytomous LCA - fits latent class models with categorical indicators
```

Section 2: Data Import and Variable Selection

```
# Import raw survey data collected via Google Forms (Bulgaria, Jan-Mar 2024)
raw <- read_excel("dataset for project.xlsx")

# Define the 12 credibility indicators used as LCA input variables
# These capture three Source Credibility Theory dimensions:
#   Q12_1-Q12_4: Attractiveness (stylish, attractive, good-looking, sexy)
```

```r
#   Q13_1-Q13_4: Expertise (knowledgeable, competent, experienced)
#   Q14_1-Q14_4: Trustworthiness (honest, authentic, trustworthy, earnest)
indicators <- c("Q12_1","Q12_2","Q12_3","Q12_4",
                "Q13_1","Q13_2","Q13_3","Q13_4",
                "Q14_1","Q14_2","Q14_3","Q14_4")

# Q17: Distal outcome - purchase frequency (recoded to binary later)
distal <- "Q17"

# Combine for full variable list used in analysis
vars <- c(indicators, distal)

# Subset to analysis variables only
df <- as.data.frame(raw[, vars])
```

Section 3: Data Recoding

```r
# LCA with poLCA requires integer-coded categorical indicators starting at 1.
# Original 5-point Likert scale responses are collapsed into 3 categories
# to reduce sparsity and improve model stability:
#   1 = Disagree  (Strongly Disagree + Disagree)
#   2 = Neutral   (Neither Agree nor Disagree)
#   3 = Agree     (Agree + Strongly Agree)

map_3cat <- c(
  "Strongly disagree"          = 1,
  "Disagree"                   = 1,
  "Neither agree nor disagree" = 2,
  "Agree"                      = 3,
  "Strongly agree"             = 3
)

# Apply recoding to all 12 credibility indicators and Q17
for (v in vars) {
  if (!is.numeric(df[[v]])) {
    df[[v]] <- unname(map_3cat[as.character(df[[v]])])
  }
  df[[v]] <- as.integer(df[[v]])
}

# Verify recoding - each variable should have values 1, 2, 3 only
lapply(df[indicators], unique)
```

```
## $Q12_1
## [1] 3 2 1
##
## $Q12_2
## [1] 2 3 1
##
```

```
## $Q12_3
## [1] 3 1 2
##
## $Q12_4
## [1] 2 1 3
##
## $Q13_1
## [1] 2 3 1
##
## $Q13_2
## [1] 2 3 1
##
## $Q13_3
## [1] 1 3 2
##
## $Q13_4
## [1] 1 2 3
##
## $Q14_1
## [1] 1 2 3
##
## $Q14_2
## [1] 1 2 3
##
## $Q14_3
## [1] 1 2 3
##
## $Q14_4
## [1] 2 3 1
```

```r
# Save cleaned analysis dataset
d <- df
```

Section 4: Descriptive Statistics

```r
# Frequency tables for all 12 credibility indicators and Q17
# These show the distribution of Disagree/Neutral/Agree responses
cat("\n=== Frequencies (3-category coding) ===\n")
```

```
##
## === Frequencies (3-category coding) ===
```

```r
for (v in vars) {
  cat("\n", v, "\n", sep = "")
  tab <- table(df[[v]], useNA = "no")
  print(tab)
  # Percentage distribution
  print(round(prop.table(tab) * 100, 2))
}
```

```
##
## Q12_1
##
##   1   2   3
## 115 149 112
##
##     1     2     3
## 30.59 39.63 29.79
##
## Q12_2
##
##   1   2   3
## 104 135 137
##
##     1     2     3
## 27.66 35.90 36.44
##
## Q12_3
##
##   1   2   3
##  65 106 205
##
##     1     2     3
## 17.29 28.19 54.52
##
## Q12_4
##
##   1   2   3
##  99 192  85
##
##     1     2     3
## 26.33 51.06 22.61
##
## Q13_1
##
##   1   2   3
## 191 109  76
##
##     1     2     3
## 50.80 28.99 20.21
##
## Q13_2
##
##   1   2   3
## 203 120  53
##
##     1     2     3
## 53.99 31.91 14.10
##
## Q13_3
##
##   1   2   3
## 245 102  29
##
```

```
##      1      2      3
## 65.16 27.13  7.71
##
## Q13_4
##
##   1   2   3
## 173 134  69
##
##      1      2      3
## 46.01 35.64 18.35
##
## Q14_1
##
##   1   2   3
## 235 119  22
##
##      1      2      3
## 62.50 31.65  5.85
##
## Q14_2
##
##   1   2   3
## 222 131  23
##
##      1      2      3
## 59.04 34.84  6.12
##
## Q14_3
##
##   1   2   3
## 216 119  41
##
##      1      2      3
## 57.45 31.65 10.90
##
## Q14_4
##
##   1   2   3
## 179 147  50
##
##      1      2      3
## 47.61 39.10 13.30
##
## Q17
##
##   1   2   3   4   5
## 175 111  67  20   3
##
##      1      2      3      4      5
## 46.54 29.52 17.82  5.32  0.80
```

```
# Summary table of mean and SD for all analysis variables
# Mean closer to 1 = more Disagree; closer to 3 = more Agree
desc_main <- data.frame(
```

```r
  Variable = vars,
  Mean = sapply(df[vars], function(x) round(mean(x, na.rm = TRUE), 2)),
  SD   = sapply(df[vars], function(x) round(sd(x,   na.rm = TRUE), 2))
)
print(desc_main)
```

```
##       Variable Mean   SD
## Q12_1    Q12_1 1.99 0.78
## Q12_2    Q12_2 2.09 0.80
## Q12_3    Q12_3 2.37 0.76
## Q12_4    Q12_4 1.96 0.70
## Q13_1    Q13_1 1.69 0.79
## Q13_2    Q13_2 1.60 0.72
## Q13_3    Q13_3 1.43 0.63
## Q13_4    Q13_4 1.72 0.75
## Q14_1    Q14_1 1.43 0.60
## Q14_2    Q14_2 1.47 0.61
## Q14_3    Q14_3 1.53 0.68
## Q14_4    Q14_4 1.66 0.70
## Q17        Q17 1.84 0.95
```

```r
# KEY FINDING - Descriptive Patterns:
# Attractiveness items (Q12): mean 1.96-2.37 - moderate agreement
# Expertise items (Q13):      mean 1.43-1.72 - generally low agreement
# Trustworthiness items (Q14):mean 1.43-1.47 - lowest scores overall
# Overall: Bulgarian respondents are skeptical of influencer credibility,
# particularly on expertise and trustworthiness dimensions.
```

Section 5: Latent Class Model Estimation (2-Class vs 3-Class)

```r
# Build the LCA formula: all 12 indicators regressed on a constant (no covariates)
# This estimates class membership based on response patterns alone
f <- as.formula(paste("cbind(", paste(indicators, collapse = ", "), ") ~ 1"))

# Set seed for reproducibility - ensures same starting values across runs
set.seed(2025)

# Fit 2-class model
# nrep = 20 random starting values to avoid local maxima
# maxiter = 2000 to ensure convergence
m2 <- poLCA(f, data = d, nclass = 2, nrep = 20, maxiter = 2000, verbose = FALSE)

# Fit 3-class model using the same settings
m3 <- poLCA(f, data = d, nclass = 3, nrep = 20, maxiter = 2000, verbose = FALSE)
```

Section 6: Model Fit Comparison

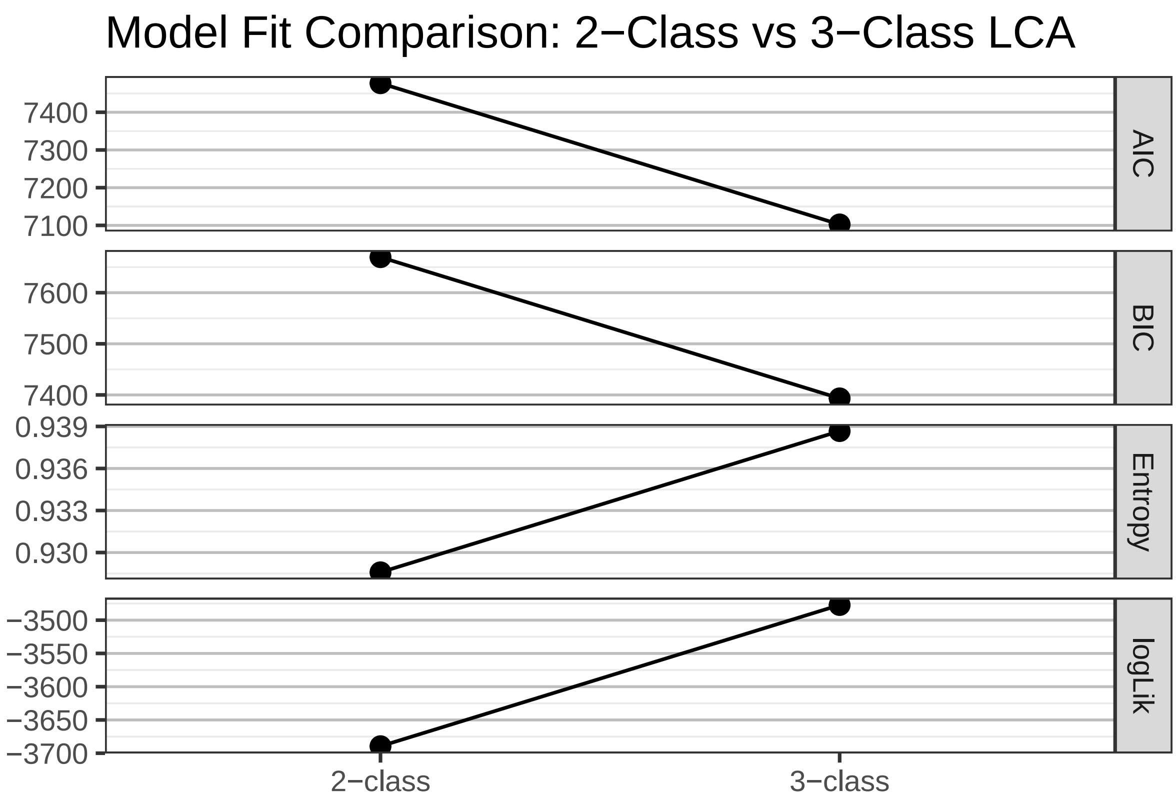
```r
# Entropy function - measures how clearly respondents are assigned to classes
# Range: 0 (no separation) to 1 (perfect separation)
# Values above 0.80 indicate good class separation (Celeux & Soromenho, 1996)
entropy <- function(posterior) {
  pp <- posterior
  pp[pp == 0] <- 1e-12     # Avoid log(0) errors
  1 - (-sum(pp * log(pp)) / (nrow(pp) * log(ncol(pp))))
}

# Compile fit statistics for both models
# Lower AIC/BIC = better fit; higher entropy = clearer class separation
cmp <- data.frame(
  Model   = c("2-class", "3-class"),
  logLik  = c(m2$llik, m3$llik),
  AIC     = c(m2$aic,  m3$aic),
  BIC     = c(m2$bic,  m3$bic),
  Entropy = c(entropy(m2$posterior), entropy(m3$posterior))
)
print(cmp)
```

```
##     Model    logLik      AIC      BIC   Entropy
## 1 2-class -3689.494 7476.988 7669.538 0.9285889
## 2 3-class -3477.255 7102.510 7393.300 0.9386682
```

```r
# KEY FINDING - Model Fit:
# 3-class model outperforms 2-class on all criteria:
#   AIC: 7,102.51 vs 7,476.99 (lower is better)
#   BIC: 7,393.30 vs 7,669.54 (lower is better)
#   Entropy: ~0.94 for both (high - clear class separation)
# The 3-class solution is preferred.

# Visualize fit statistics across model solutions
cmp$Model <- factor(cmp$Model, levels = c("2-class", "3-class"))

cmp_long <- cmp %>%
  pivot_longer(cols = c(logLik, AIC, BIC, Entropy),
               names_to  = "Criterion",
               values_to = "Value")

ggplot(cmp_long) +
  geom_point(aes(x = Model, y = Value), size = 3) +
  geom_line(aes(x = Model, y = Value, group = 1)) +
  facet_grid(Criterion ~ ., scales = "free") +
  theme_bw(base_size = 14) +
  labs(title = "Model Fit Comparison: 2-Class vs 3-Class LCA",
       x = "", y = "") +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.major.y = element_line(colour = "grey", size = 0.5))
```

```
## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
```

```
## i Please use the `linewidth` argument instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Section 7: Bootstrap Likelihood Ratio Test (BLRT)

```
# The BLRT formally tests whether K classes fit significantly better than K-1
# H0: K-1 classes are sufficient
# H1: K classes provide significantly better fit
# Significant p-value (< .05) supports retaining the larger model

blrt_poLCA <- function(data, indicators, k,
                       B = 200, nrep = 20, maxiter = 2000,
                       seed = 1234, verbose = TRUE) {
  stopifnot(k >= 2)
  set.seed(seed)

  # Use complete cases only - ensures clean integer-coded indicators
  idx <- stats::complete.cases(data[, indicators, drop = FALSE])
  dat <- data[idx, indicators, drop = FALSE]
```

```r
for (v in indicators) dat[[v]] <- as.integer(dat[[v]])

f <- as.formula(paste("cbind(", paste(indicators, collapse = ", "), ") ~ 1"))

# Wrap model fitting in tryCatch to handle non-convergence gracefully
safe_fit <- function(form, dat, k, nrep, maxiter) {
  tryCatch(
    poLCA(form, dat, nclass = k, nrep = nrep,
          maxiter = maxiter, verbose = FALSE),
    error = function(e) NULL
  )
}

# Compute observed likelihood ratio statistic on real data
fit_km1 <- safe_fit(f, dat, k - 1, nrep, maxiter)
fit_k   <- safe_fit(f, dat, k,     nrep, maxiter)
if (is.null(fit_km1) || is.null(fit_k)) {
  stop("Observed models did not converge for k-1 or k.")
}
LR_obs <- 2 * (fit_k$llik - fit_km1$llik)
if (verbose) cat(sprintf("Observed LR (k=%d vs k-1=%d): %.3f\n",
                          k, k - 1, LR_obs))

# Generate B bootstrap samples from the (K-1)-class null model
# Each sample is simulated from the null model's estimated parameters
sim_from_model <- function(fit, n) {
  K <- length(fit$P)
  J <- length(fit$probs)
  out <- as.data.frame(matrix(NA_integer_, nrow = n, ncol = J))
  names(out) <- names(fit$probs)
  cls <- sample.int(K, size = n, replace = TRUE, prob = fit$P)
  for (j in seq_len(J)) {
    pj     <- fit$probs[[j]]
    Cj     <- ncol(pj)
    draw_j <- integer(n)
    for (kk in seq_len(K)) {
      ii <- which(cls == kk)
      if (length(ii)) {
        draw_j[ii] <- sample.int(Cj, length(ii),
                                 replace = TRUE, prob = pj[kk, ])
      }
    }
    out[[j]] <- draw_j
  }
  out
}

n       <- nrow(dat)
LR_boot <- rep(NA_real_, B)

# Fit both models to each bootstrap sample and compute LR
for (b in seq_len(B)) {
  simdat    <- sim_from_model(fit_km1, n)
```

```r
    fit_b_km1 <- safe_fit(f, simdat, k - 1, nrep, maxiter)
    fit_b_k   <- safe_fit(f, simdat, k,     nrep, maxiter)

    if (is.null(fit_b_km1) || is.null(fit_b_k)) {
      if (verbose) cat("Non-convergence in bootstrap replicate", b, "- skipping\n")
      next
    }
    LR_boot[b] <- 2 * (fit_b_k$llik - fit_b_km1$llik)

    if (verbose && (b %% max(1, round(B / 10)) == 0)) {
      cat(sprintf("BLRT progress: %d/%d\n", b, B))
    }
  }

  # p-value = proportion of bootstrap LR statistics >= observed LR
  valid <- !is.na(LR_boot)
  if (!any(valid)) stop("All bootstrap replicates failed to converge.")
  pval <- mean(LR_boot[valid] >= LR_obs)

  list(k = k, LR_obs = LR_obs, LR_boot = LR_boot,
       p_value = pval, B_used = sum(valid))
}

# Run BLRT: test whether 3-class fits better than 2-class
blrt_3v2 <- blrt_poLCA(d, indicators, k = 3,
                        B = 200, nrep = 20, maxiter = 2000, verbose = TRUE)
```

```
## Observed LR (k=3 vs k-1=2): 424.478
## BLRT progress: 20/200
## BLRT progress: 40/200
## BLRT progress: 60/200
## BLRT progress: 80/200
## BLRT progress: 100/200
## BLRT progress: 120/200
## BLRT progress: 140/200
## BLRT progress: 160/200
## BLRT progress: 180/200
## BLRT progress: 200/200
```

```r
cat(sprintf("BLRT 3 vs 2: LR = %.2f, p = %.3f, B_used = %d\n",
            blrt_3v2$LR_obs, blrt_3v2$p_value, blrt_3v2$B_used))
```

```
## BLRT 3 vs 2: LR = 424.48, p = 0.000, B_used = 200
```

```r
# Significant result (p < .001) confirms the 3-class solution is preferred
```

Section 8: Select Best Model and Extract Class Profiles

```r
# Select the model with the lower BIC (penalizes complexity more than AIC)
best <- if (cmp$BIC[1] < cmp$BIC[2]) m2 else m3   # Selects m3
K    <- ncol(best$posterior)

# Item-response probabilities: Pr(Disagree/Neutral/Agree) for each indicator per class
# These are the core output used to characterize and label each latent class
cat("\nItem-response probabilities by class\n")
```

```
##
## Item-response probabilities by class
```

```r
probs_list <- lapply(best$probs, function(x) round(x, 3))
print(probs_list)
```

```
## $Q12_1
##           Pr(1) Pr(2) Pr(3)
## class 1:  0.059 0.516 0.425
## class 2:  0.971 0.029 0.000
## class 3:  0.189 0.471 0.339
##
## $Q12_2
##           Pr(1) Pr(2) Pr(3)
## class 1:  0.034 0.425 0.541
## class 2:  0.919 0.070 0.012
## class 3:  0.167 0.436 0.397
##
## $Q12_3
##           Pr(1) Pr(2) Pr(3)
## class 1:  0.000 0.306 0.694
## class 2:  0.818 0.182 0.000
## class 3:  0.018 0.307 0.675
##
## $Q12_4
##           Pr(1) Pr(2) Pr(3)
## class 1:  0.062 0.631 0.307
## class 2:  0.831 0.169 0.000
## class 3:  0.158 0.574 0.268
##
## $Q13_1
##           Pr(1) Pr(2) Pr(3)
## class 1:  0.073 0.487 0.441
## class 2:  0.946 0.054 0.000
## class 3:  0.606 0.261 0.133
##
## $Q13_2
##           Pr(1) Pr(2) Pr(3)
## class 1:  0.076 0.599 0.324
## class 2:  0.950 0.050 0.000
## class 3:  0.668 0.250 0.082
##
## $Q13_3
##           Pr(1) Pr(2) Pr(3)
```

```
## class 1:  0.252 0.527 0.221
## class 2:  0.962 0.038 0.000
## class 3:  0.780 0.204 0.017
##
## $Q13_4
##            Pr(1) Pr(2) Pr(3)
## class 1:  0.087 0.565 0.348
## class 2:  0.939 0.045 0.016
## class 3:  0.502 0.351 0.147
##
## $Q14_1
##            Pr(1) Pr(2) Pr(3)
## class 1:  0.093 0.729 0.178
## class 2:  0.959 0.041 0.000
## class 3:  0.829 0.165 0.006
##
## $Q14_2
##            Pr(1) Pr(2) Pr(3)
## class 1:  0.032 0.789 0.179
## class 2:  0.962 0.038 0.000
## class 3:  0.795 0.194 0.011
##
## $Q14_3
##            Pr(1) Pr(2) Pr(3)
## class 1:  0.034 0.705 0.261
## class 2:  0.986 0.000 0.014
## class 3:  0.752 0.198 0.051
##
## $Q14_4
##            Pr(1) Pr(2) Pr(3)
## class 1:  0.030 0.646 0.324
## class 2:  0.926 0.074 0.000
## class 3:  0.576 0.359 0.065
```

```r
# Class proportions - estimated prevalence of each class in the population
cat("\nClass proportions\n")
```

```
##
## Class proportions
```

```r
print(round(best$P, 4))
```

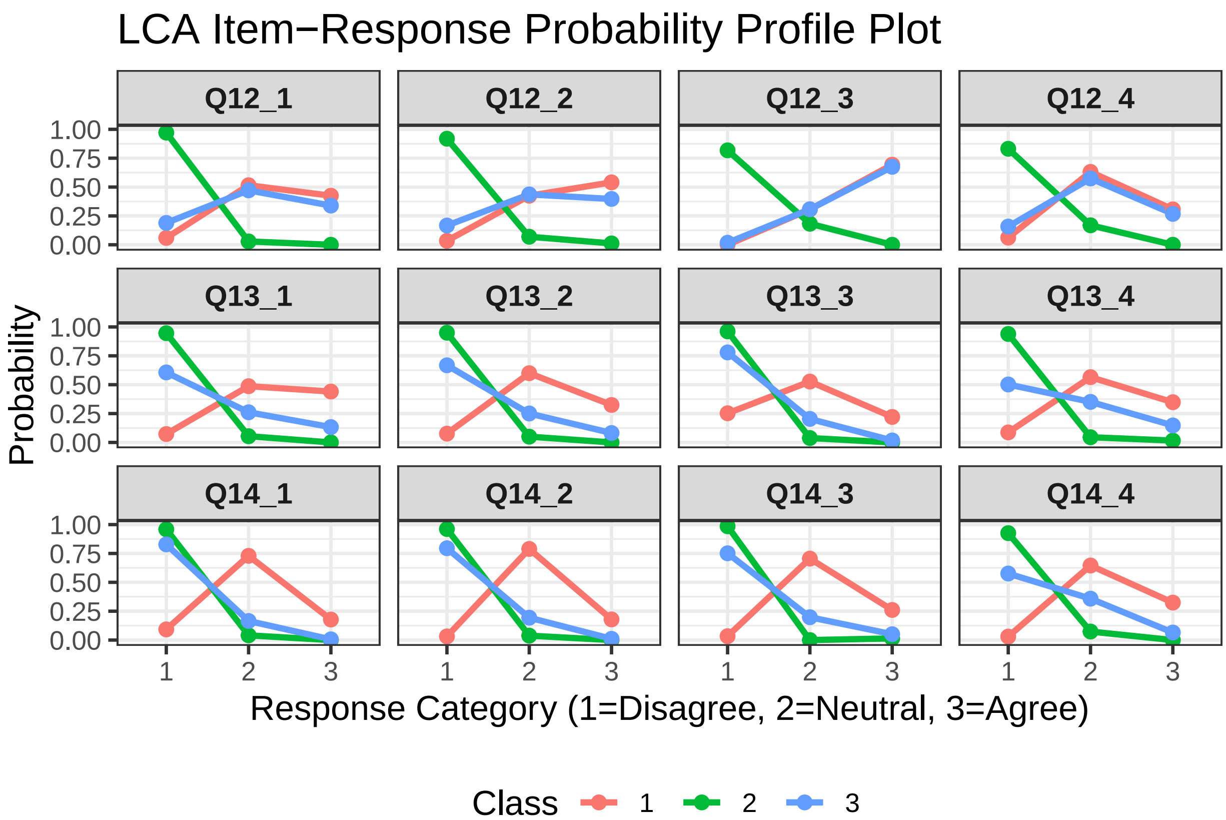
```
## [1] 0.3122 0.2010 0.4868
```

Section 9: LCA Profile Plot (Item-Level)

```r
# Visualize item-response probabilities across all 12 indicators
# Each panel shows one indicator; lines show response probability profiles per class
```

```r
probs <- m3$probs

plot_df <- lapply(names(probs), function(item) {
  df_item        <- as.data.frame(probs[[item]])
  df_item$Item   <- item
  df_item$Class  <- rownames(probs[[item]]) %>%
                    str_replace("class ", "") %>%
                    str_replace(":", "")
  df_item %>%
    pivot_longer(cols      = starts_with("Pr"),
                 names_to  = "Response",
                 values_to = "Probability") %>%
    mutate(
      Response = str_replace(Response, "Pr\\(", "") %>%
                 str_replace("\\)", ""),
      Class    = factor(Class),
      Item     = factor(Item, levels = names(probs))
    )
}) %>% bind_rows()

ggplot(plot_df, aes(x = Response, y = Probability,
                    color = Class, group = Class)) +
  geom_line(size = 1.1) +
  geom_point(size = 2) +
  facet_wrap(~ Item, ncol = 4) +
  theme_bw(base_size = 13) +
  labs(title = "LCA Item-Response Probability Profile Plot",
       x     = "Response Category (1=Disagree, 2=Neutral, 3=Agree)",
       y     = "Probability") +
  theme(strip.text      = element_text(size = 11, face = "bold"),
        legend.position = "bottom")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



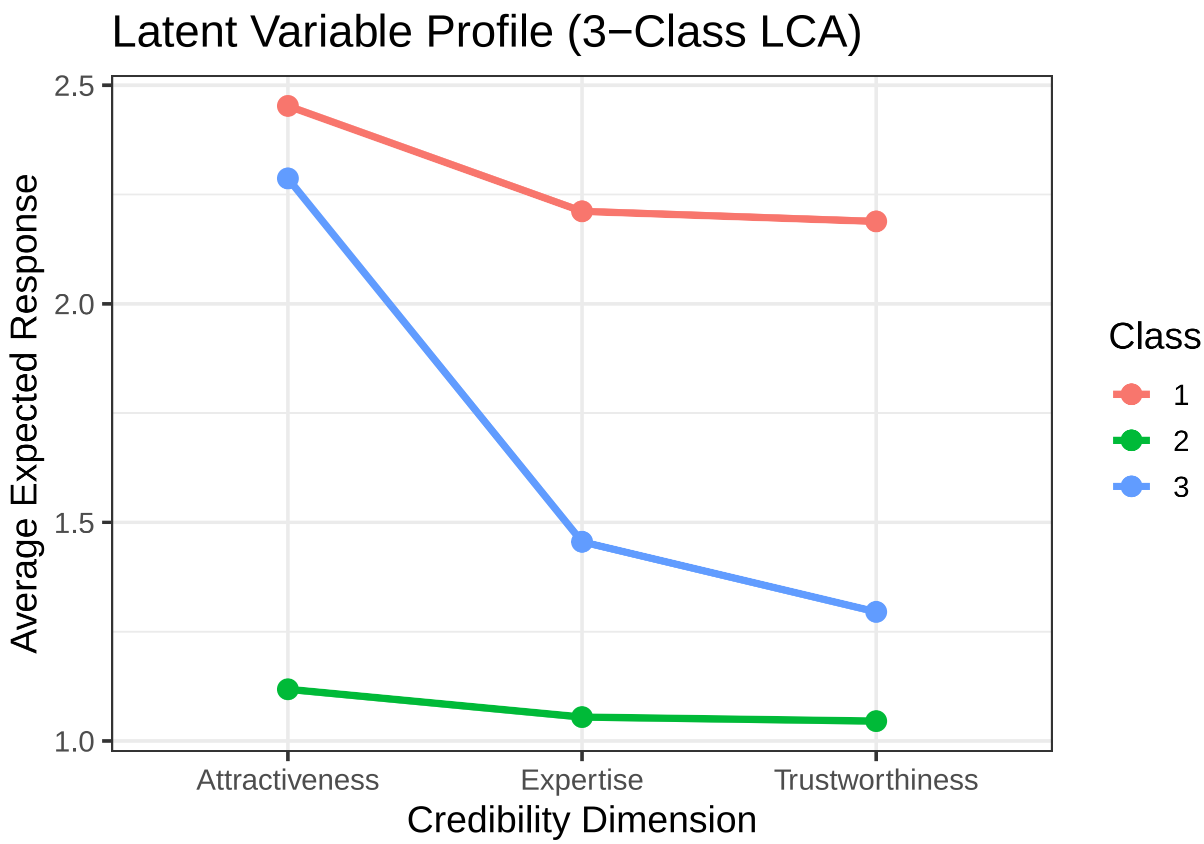
Section 10: Latent Variable Profile Plot (Dimension-Level)

```
# Compute expected response scores per latent dimension per class
# Expected value = weighted average: Pr(1)*1 + Pr(2)*2 + Pr(3)*3
# Higher scores = more favorable credibility evaluations within that dimension

lv1_items <- c("Q12_1","Q12_2","Q12_3","Q12_4")    # Attractiveness
lv2_items <- c("Q13_1","Q13_2","Q13_3","Q13_4")    # Expertise
lv3_items <- c("Q14_1","Q14_2","Q14_3","Q14_4")    # Trustworthiness

expected_value <- function(mat) {
  mat[, 1] * 1 + mat[, 2] * 2 + mat[, 3] * 3
}

irp <- m3$probs

# Compute mean expected score for each credibility dimension within each class
lv_means <- data.frame(
  Class = factor(1:3),
  Attractiveness = sapply(1:3, function(k) {
    mean(sapply(lv1_items, function(it) expected_value(irp[[it]])[k]))
  }),
```

```r
  Expertise = sapply(1:3, function(k) {
    mean(sapply(lv2_items, function(it) expected_value(irp[[it]])[k]))
  }),
  Trustworthiness = sapply(1:3, function(k) {
    mean(sapply(lv3_items, function(it) expected_value(irp[[it]])[k]))
  })
)

print(lv_means)
```

```
##   Class Attractiveness Expertise Trustworthiness
## 1     1       2.452647  2.211562        2.188464
## 2     2       1.118232  1.054627        1.045413
## 3     3       2.286841  1.455622        1.295313
```

```r
# Convert to long format and plot dimension-level profiles
lv_long <- pivot_longer(lv_means,
                        cols      = c("Attractiveness","Expertise","Trustworthiness"),
                        names_to  = "LatentVariable",
                        values_to = "Score")

ggplot(lv_long, aes(x = LatentVariable, y = Score,
                    color = Class, group = Class)) +
  geom_line(size = 1.3) +
  geom_point(size = 3) +
  theme_bw(base_size = 14) +
  labs(title = "Latent Variable Profile (3-Class LCA)",
       x     = "Credibility Dimension",
       y     = "Average Expected Response")
```



Section 11: Class Labeling (High-Trust, Neutral, Low-Trust)

```
# Assign descriptive labels to each latent class using two criteria:
#   (1) Average Pr(Agree) across all 12 indicators
#   (2) Number of indicators where Neutral is the dominant response category
#
# Labeling rules:
#   High-Trust:    Avg Pr(Agree) >= 0.35
#   Neutral:       Neutral response dominates on >= 3 indicators
#   Low-Trust:     Neither condition met (Disagree dominates)

K <- ncol(best$posterior)

agree_probs <- sapply(seq_len(K), function(k) {
  mean(sapply(best$probs, function(M) M[k, "Pr(3)"]))
})

neutral_dom_count <- sapply(seq_len(K), function(k) {
  sum(sapply(best$probs, function(M) {
    which.max(M[k, ]) == which(colnames(M) == "Pr(2)")
  }))
})
```

```r
lowtrust_dom_count <- sapply(seq_len(K), function(k) {
  sum(sapply(best$probs, function(M) {
    which.max(M[k, ]) == which(colnames(M) == "Pr(1)")
  }))
})

hightrust_dom_count <- sapply(seq_len(K), function(k) {
  sum(sapply(best$probs, function(M) {
    which.max(M[k, ]) == which(colnames(M) == "Pr(3)")
  }))
})

labels <- sapply(seq_len(K), function(k) {
  if (agree_probs[k] >= 0.35)       "High-Trust"
  else if (neutral_dom_count[k] >= 3) "Moderate/Neutral"
  else                               "Low-Trust"
})

# Summary table: class proportions and labels
data.frame(
  Class            = paste0("Class", 1:K),
  Avg_Pr_Agree     = round(agree_probs, 3),
  NeutralDominance = neutral_dom_count,
  Label            = labels,
  Proportion       = round(best$P, 3)
)
```

```
##    Class Avg_Pr_Agree NeutralDominance            Label Proportion
## 1 Class1        0.354               10       High-Trust      0.312
## 2 Class2        0.003                0        Low-Trust      0.201
## 3 Class3        0.182                3 Moderate/Neutral      0.487
```

```r
# KEY FINDING - Class Profiles:
# Class 1 (31.2%): High-Trust - highest Avg Pr(Agree) ~0.35, reserved but positive
# Class 2 (20.1%): Low-Trust  - Pr(Disagree) > 0.80 on all 12 indicators
# Class 3 (48.7%): Neutral    - largest class; moderate on attractiveness,
#                               skeptical on expertise and trustworthiness
```

Section 12: Assign Class Membership to Respondents

```r
# Assign each respondent to their modal class (highest posterior probability)
# This is the "most likely class" assignment used in the distal outcome analysis
idx        <- complete.cases(d[, indicators])
d$class    <- NA_integer_
d$class[idx] <- apply(best$posterior, 1, which.max)
d$class    <- factor(d$class,
                     levels = 1:K,
                     labels = paste0("Class", 1:K))
```

```r
# Verify class distribution
cat("\nClass membership counts\n")
```

```
##
## Class membership counts
```

```r
print(table(d$class, useNA = "ifany"))
```

```
##
## Class1 Class2 Class3
##    119     76    181
```

Section 13: Recode Distal Outcome (Purchase Behavior)

```r
# Q17 original scale: 1=Never, 2=Rarely, 3=Sometimes, 4=Often, 5=Very Often
# Recoded to binary for logistic regression:
#   0 = Never purchased (Q17 = 1)
#   1 = Ever purchased  (Q17 = 2, 3, 4, or 5)
# This ensures sufficient cell sizes for regression analysis

cat("\nOriginal Q17 distribution\n")
```

```
##
## Original Q17 distribution
```

```r
print(table(d$Q17))
```

```
##
##   1   2   3   4   5
## 175 111  67  20   3
```

```r
print(round(prop.table(table(d$Q17)) * 100, 2))
```

```
##
##     1     2     3     4     5
## 46.54 29.52 17.82  5.32  0.80
```

```r
d$Q17_2cat                    <- d$Q17
d$Q17_2cat[d$Q17 == 1]             <- 0    # Never
d$Q17_2cat[d$Q17 %in% c(2,3,4,5)] <- 1    # Ever purchased

cat("\nRecoded Q17 (binary)\n")
```

```
##
## Recoded Q17 (binary)
```

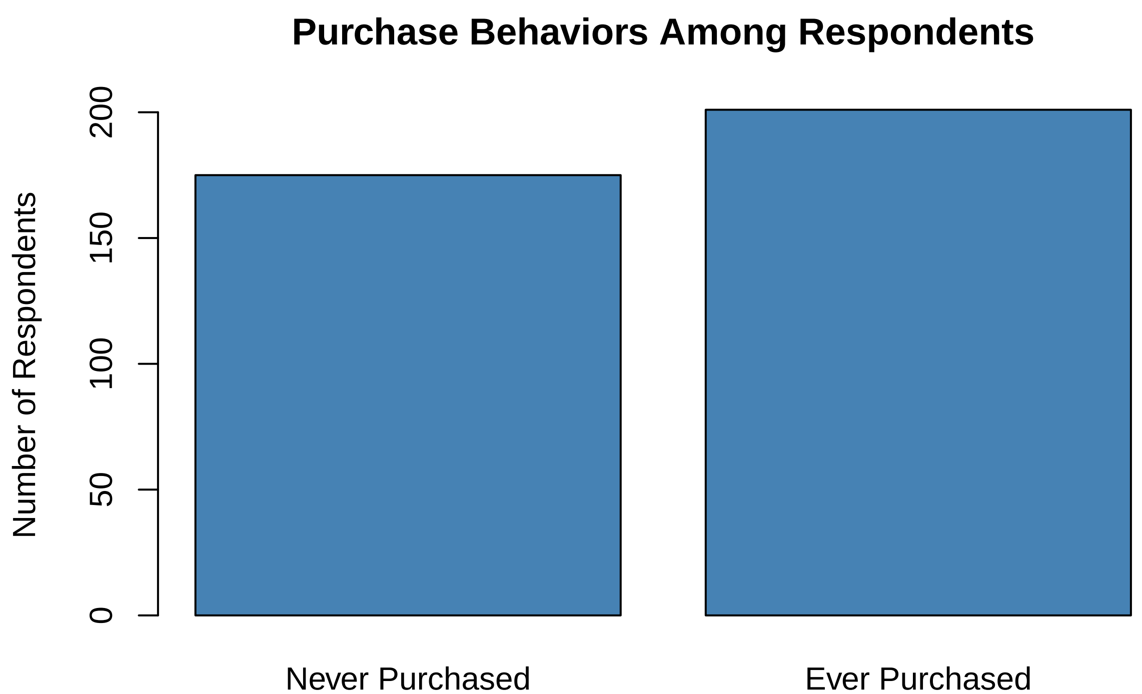
```r
print(table(d$Q17_2cat))
```

```
##
##   0   1
## 175 201
```

```r
print(round(prop.table(table(d$Q17_2cat)) * 100, 1))
```

```
##
##    0    1
## 46.5 53.5
```

```r
# Bar chart of purchase behavior distribution
barplot(table(d$Q17_2cat),
        names.arg = c("Never Purchased", "Ever Purchased"),
        col       = "steelblue",
        main      = "Purchase Behaviors Among Respondents",
        ylab      = "Number of Respondents")
```



Section 14: Logistic Regression — Class Membership Predicting Purchase Behavior

```r
# Fit a binary logistic regression model predicting ever-purchase (Q17_2cat)
# from latent class membership (Class1 = High-Trust as reference category)
# This is a two-step approach: class assignments are treated as fixed predictors

m4 <- glm(Q17_2cat ~ factor(class), data = d, family = "binomial")
summary(m4)
```

```
##
## Call:
## glm(formula = Q17_2cat ~ factor(class), family = "binomial",
##     data = d)
##
## Coefficients:
##                     Estimate Std. Error z value Pr(>|z|)
## (Intercept)           0.8353     0.1996   4.186 2.84e-05 ***
## factor(class)Class2  -1.7333     0.3222  -5.380 7.46e-08 ***
## factor(class)Class3  -0.7136     0.2490  -2.866  0.00416 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 519.45  on 375  degrees of freedom
## Residual deviance: 487.60  on 373  degrees of freedom
## AIC: 493.6
##
## Number of Fisher Scoring iterations: 4
```

```r
# Convert log-odds to predicted probabilities for each class
coefs <- coef(m4)

# Class 1 (baseline/reference)
logodds_c1 <- coefs[1]
prob_c1    <- plogis(logodds_c1)

# Class 2 (vs Class 1)
logodds_c2 <- coefs[1] + coefs["factor(class)Class2"]
prob_c2    <- plogis(logodds_c2)

# Class 3 (vs Class 1)
logodds_c3 <- coefs[1] + coefs["factor(class)Class3"]
prob_c3    <- plogis(logodds_c3)

# Summary: predicted probability of purchase by class
results <- data.frame(
  Class       = c("Class1 (High-Trust)", "Class2 (Low-Trust)", "Class3 (Neutral)"),
  LogOdds     = round(c(logodds_c1, logodds_c2, logodds_c3), 3),
  Probability = round(c(prob_c1,    prob_c2,    prob_c3),    3)
)
print(results)
```

```
##                   Class LogOdds Probability
## 1 Class1 (High-Trust)   0.835       0.697
## 2  Class2 (Low-Trust)  -0.898       0.289
## 3    Class3 (Neutral)   0.122       0.530
```

```
# KEY FINDING - Purchase Behavior by Class:
# High-Trust class (Class 1): ~70% predicted probability of purchase
#   - Respondents who perceive influencers as credible across all three dimensions
#     are most likely to act on influencer recommendations.
#
# Neutral class (Class 3): ~53% predicted probability of purchase
#   - Moderate credibility perceptions translate to moderate purchase likelihood,
#     slightly above the sample average (~53% ever purchased overall).
#
# Low-Trust class (Class 2): ~29% predicted probability of purchase
#   - Respondents who consistently disagree with influencer credibility statements
#     are substantially less likely to purchase, nearly half the rate of High-Trust.
#
# This gradient (70% > 53% > 29%) is consistent with Source Credibility Theory:
# perceived credibility is a meaningful predictor of consumer purchase behavior.
```

Section 15: Summary of Key Findings

Model Selection

The 3-class LCA model was selected over the 2-class solution based on four fit criteria. The 3-class model produced a lower AIC (7,102.51 vs 7,476.99) and lower BIC (7,393.30 vs 7,669.54), indicating better fit with appropriate complexity penalization. Entropy was high for both models (~0.94), confirming clear class separation regardless of solution size. A Bootstrap Likelihood Ratio Test (BLRT, B = 200) formally confirmed that 3 classes fit significantly better than 2 (LR = 424.48, $p < .001$).

Identified Consumer Profiles

Three distinct consumer profiles were identified based on response patterns across 12 credibility indicators:

- **High-Trust (Class 1, 31.2%):** Respondents with moderate-to-high agreement across attractiveness, expertise, and trustworthiness indicators. This group showed the highest average Pr(Agree) (~0.35) and strong neutral tendencies on trustworthiness items, reflecting a reserved but positive credibility perception.
- **Low-Trust (Class 2, 20.1%):** The most homogeneous and distinctive class. All 12 indicators showed Pr(Disagree) > 0.80, indicating near-universal rejection of influencer credibility across all three dimensions. Average Pr(Agree) was only 0.003.
- **Neutral (Class 3, 48.7%):** The largest class, characterized by selective perception — moderate agreement on attractiveness items but substantial disagreement on expertise and trustworthiness. This group represents the majority of Bulgarian social media users in this sample.

Key Credibility Dimension Findings

Trustworthiness items (Q14_1–Q14_4) showed the largest separation across classes, identifying trustworthiness as the primary dimension driving latent segmentation. Attractiveness was evaluated more favorably than expertise or trustworthiness across all three classes, suggesting consumers find influencers visually appealing even when skeptical of their knowledge or honesty.

Purchase Behavior Prediction

Logistic regression confirmed a clear relationship between credibility class membership and purchase behavior:

Class	Predicted Purchase Probability
High-Trust	~70%
Neutral	~53%
Low-Trust	~29%

This gradient supports the theoretical expectation that higher perceived credibility increases susceptibility to influencer-driven purchasing decisions. The findings suggest that trustworthiness — the dimension most strongly separating classes — may be the most important lever for influencer marketing effectiveness in this population.

Limitations

- Data were collected under simple random sampling assumptions; no survey weights were applied, which may affect generalizability.
- Class assignments were treated as fixed in the logistic regression (two-step approach), which does not account for classification uncertainty. Methods such as BCH in Mplus would better handle this variability.
- Cross-sectional design limits causal inference between credibility perceptions and purchase behavior.